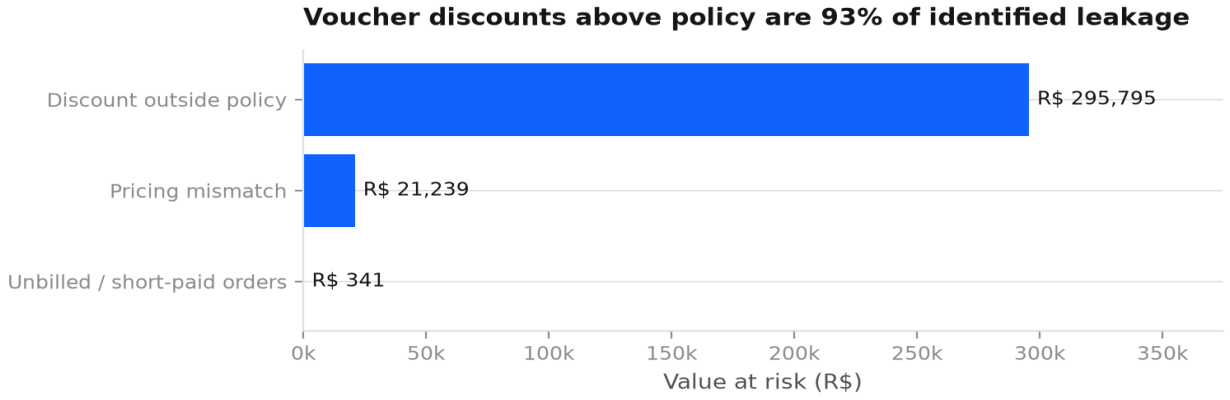
Revenue Assurance and Leakage Review

Olist Brazilian e-commerce orders, 2016 to 2018 | Source: Olist public dataset (Kaggle, CC BY-NC-SA 4.0) | Currency: Brazilian real (R$)

1. Scope and result

Reconciled 96,470 delivered orders (R$ 15,418,395 billed) against order lines, delivery records and payments, and tested three leakage types. Identified value at risk is **R$ 317,375**, 2.1% of delivered billed value. Billed value and cash collected agree to within R$ 2,688 on delivered orders, so the leakage sits in discounts and prices, not in missing payments.

Rank	Leakage type	Exceptions	Value at risk	Share of leakage	% of billed
1	Discount outside policy	3,600	R$ 295,795	93.2%	1.92%
2	Pricing mismatch	1,308	R$ 21,239	6.7%	0.14%
3	Unbilled / short-paid orders	18	R$ 341	0.1%	<0.01%
	Total	**4,926**	**R$ 317,375**	100%	2.06%



2. Findings

F1 Discounts outside policy (R$ 295,795). 3,600 of 3,679 voucher orders funded more than 10% of the order with a voucher. The median voucher order is 93% voucher-funded and 1,565 are fully voucher-funded, so the result moves little with the cap: R$ 319,071 at 5%, R$ 251,527 at 20%. Value at risk is the voucher amount above the cap.

F2 Pricing mismatches (R$ 21,239). 1,308 item lines were billed more than 10% below the list price. A further 1,483 lines were billed more than 10% above list (R$ 31,278); that is a customer-dispute exposure and is not counted as leakage.

F3 Unbilled or short-paid orders (R$ 341). 18 delivered orders were paid less than billed; one delivered order has no payment record. Immaterial in value, but cheap to control with a daily match.

Observation, not leakage. 1,239 cancelled or unavailable orders carry payments of R$ 271,114, and most unavailable orders have no order lines. The data holds no refund records, so whether this cash was returned cannot be confirmed. Also 8 orders marked delivered have no delivery date.

3. Reconciliation by order status

Order status	Orders	Billed (R$)	Paid (R$)	Paid minus billed
delivered	96,478	15,419,774	15,422,462	2,688
shipped	1,107	177,129	177,214	85
canceled	625	105,886	143,256	37,370
unavailable	609	2,140	126,480	124,339
invoiced	314	68,989	69,138	149
processing	301	69,394	69,394	0
created	5	0	688	688
approved	2	241	241	0
Total	**99,441**	**15,843,553**	**16,008,872**	**165,319**

Billed = item price plus freight. Totals tie to the source tables.

4. Method and assumptions

Olist has no price list, discount policy or invoice table, so the policy parameters below are set by the reviewer. Each can be changed in `src/analysis.py` and the figures rebuild.

Item	Setting
Billed value	Item price plus freight per order (the order lines are the billing record)
List price	Median unit price of the product in the purchase quarter, for products with at least 5 sales (45,997 of 110,189 delivered lines, 42%, are testable)
Price tolerance	Plus or minus 10% of list. Sensitivity: 5% gives R$ 30,525, 20% gives R$ 6,929
Short-pay tolerance	R$ 1.00 per order (instalment rounding)
Voucher cap	10% of billed value without approval; leakage is the excess
No double count	Steps run in sequence: price variance is measured on item price, short-pay on billed less paid, voucher excess on the voucher payment. They draw on different amounts.

Limits: the list price is observed, not contractual, and a marketplace seller can price legitimately below the quarter median. Credit notes and reason codes do not exist in the data and are not tested. Findings show where to look; they are not confirmed losses.